Metric Framework and First Variation of the Volume-Radius Foliation (Route δ, Plan 2, Building Block MetricFramework)

Abstract

We set up the metric ambient space $(\mathcal{X}, d_{\mathcal{X}})$ of L^1-equivalence classes modulo translations, parametrize the torsion super-level sets by the volume radius, and prove the metric first-variation estimate (T) for the rescaled curve $\rho \mapsto [F_\rho] \in \mathcal{X}$ in finite-perimeter generality, with no graph parametrization of $\partial^* E_\rho$. The proof has three ingredients: (i) the Sobolev–Sard theorem of Figalli–Maggi 2011 (Appendix A, Theorem A.1), which makes the normal velocity $V_\rho = -t'(\rho)/|\nabla u|$ pointwise defined $\mathcal{H}^{n-1}$-a.e. on $\partial^* E_\rho$ for a.e. ρ; (ii) the moving-region first-variation calculus for smooth nested flows, extended to finite-perimeter levels by BV/coarea approximation, perimeter lower semicontinuity, Reshetnyak lower semicontinuity for linear-growth integrands, and the Ambrosio–Gigli–Savaré lower semicontinuity of the metric derivative; (iii) the translation-volume identity for finite-perimeter sets, which shows that the translation infimum on the right-hand side of (T) is precisely the metric-quotient cost. The centre field z_ρ enters as an arbitrary Borel measurable map; in the repaired downstream chain it is instantiated in Building Block WeightedMetricTrace as a levelwise Fusco–Julin oscillation centre on small-isoperimetric-defect levels and as an arbitrary bounded centre on the complementary bad set. We state honestly that the pointwise-in-ρ squared form is conditional on a per-ρ radial trace that is open in the no-graph setting; the paper downstream uses only the integrated good/bad form proved in Building Block WeightedMetricTrace.

1 Standing hypotheses

Let $n \geq 2$ and let $\Omega \subset \mathbb{R}^n$ be a bounded open set with $|\Omega| = \omega_n = |B_1|$ and $\Omega \subset B_R$ (without loss of generality $R \geq 2$ after rescaling). Let $u \in H^1_0(\Omega) \cap L^\infty(\Omega)$ be the torsion potential, the unique variational solution of

$$-\Delta u = 1 \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega.$$

The L^∞ bound follows from comparison with the torsion function of B_R; interior elliptic regularity gives $u \in W^{2,p}_{\mathrm{loc}}(\Omega)$ for every $p < \infty$. Fix $\rho_* \in (0, 1)$. The volume-radius parametrization is

$$E_\rho := \{u > t(\rho)\}, \qquad |E_\rho| = \omega_n \rho^n, \qquad \rho \in [\rho_*, 1], \tag{1}$$

with $\rho \mapsto t(\rho)$ absolutely continuous on $[\rho_*, 1]$ and $t'(\rho) \leq 0$ a.e. (by monotonicity of $\rho \mapsto t(\rho)$ and the volume identity $|E_\rho| = \omega_n \rho^n$; see also [3] Thm. 13.1 and [2] Thm. 3.40). For a.e. ρ, E_ρ has finite perimeter, reduced boundary $\partial^* E_\rho$, outer unit normal ν_ρ, and $P(E_\rho) = \mathcal{H}^{n-1}(\partial^* E_\rho)$. We write $P_\rho := P(B_\rho) = n\omega_n \rho^{n-1}$.

2 The metric space $\mathcal{X}$ and the rescaled curve

Definition 2.1 (L^1 modulo translations). Let $\mathfrak{M}$ denote the set of $\mathcal{L}^n$-equivalence classes of characteristic functions $\mathbf{1}_A$ with $|A| < \infty$, equipped with the L^1 distance $d_{L^1}(\mathbf{1}_A, \mathbf{1}_C) = |A \Delta C|$.

The group $\mathbb{R}^n$ acts on $\mathfrak{M}$ by translation. The metric space $(\mathcal{X}, d_{\mathcal{X}})$ is the quotient

$$\mathcal{X} := \mathfrak{M}/\mathbb{R}^n, \qquad d_{\mathcal{X}}([A],[C]) := \inf_{a\in\mathbb{R}^n} |A\Delta(C+a)|. \tag{2}$$

Lemma 2.2 (Attainment of the translation infimum)**.** *If $A, C \subset \mathbb{R}^n$ have finite measure and are both contained in some ball B_M, the infimum in* (2) *is attained at some $a_* \in \overline{B_{2M}}$.*

Proof. For $|a| \geq 2M$, $(C+a)\cap A = \emptyset$ and $|A\Delta(C+a)| = |A| + |C|$, which exceeds $|A| + |C| - |A\cap C| = |A\Delta C| + |A\cap C| \geq |A\Delta C|$, so the infimum is unchanged if we restrict to $a \in \overline{B_{2M}}$. The map $a \mapsto |A\Delta(C+a)|$ is continuous on $\mathbb{R}^n$ (by dominated convergence), hence attains its minimum on the compact set $\overline{B_{2M}}$. □

Definition 2.3 (Rescaled curve)**.** For $\rho \in [\rho_*, 1]$, define the volume-rescaled level set and its class

$$F_\rho := \rho^{-1}E_\rho \subset \mathbb{R}^n, \qquad [F_\rho] \in \mathcal{X}.$$

By construction $|F_\rho| = \omega_n = |B_1|$ for all ρ. Since $E_\rho \subset B_R$, we have $F_\rho \subset B_{R/\rho_*}$ uniformly in $\rho \in [\rho_*, 1]$, and Lemma 2.2 applies to all pairs $([F_{\rho_1}], [F_{\rho_2}])$.

Definition 2.4 (Metric derivative; Ambrosio–Gigli–Savaré [1, Thm. 1.1.2])**.** A curve $\gamma : I \to \mathcal{X}$ is absolutely continuous if there is $m \in L^1(I)$ with $d_{\mathcal{X}}(\gamma(s), \gamma(\sigma)) \leq \int_\sigma^s m$ for $\sigma \leq s$ in I. The metric derivative

$$|\dot\gamma|(\rho) := \lim_{h\to 0} \frac{d_{\mathcal{X}}(\gamma(\rho+h), \gamma(\rho))}{|h|}$$

exists for a.e. $\rho \in I$ and is the smallest admissible m.

3 The normal velocity V_ρ and the Sobolev–Sard step

For a.e. $\rho \in [\rho_*, 1]$, set

$$V_\rho(x) := \frac{-t'(\rho)}{|\nabla u(x)|} \qquad \text{for } x \in \partial^* E_\rho,\ |\nabla u|(x) > 0. \tag{3}$$

Proposition 3.1 (Sobolev–Sard for $W^{2,p}_{\mathrm{loc}}$)**.** *Let $p > n$ (via Morrey embedding $W^{2,p} \hookrightarrow C^{1,\alpha}$, $\alpha = 1 - n/p$). Then*

$$\mathcal{H}^{n-1}\big(\partial^* E_\rho \cap \{|\nabla u| = 0\}\big) = 0 \qquad \textit{for a.e. } \rho \in [\rho_*, 1]. \tag{4}$$

Proof. By Stampacchia, $u \in W^{2,p}_{\mathrm{loc}}(\Omega)$ for every $p < \infty$; fix any $p > n$. The Sobolev–Sard theorem of Figalli and Maggi [4, Appendix A, Thm. A.1] states that for $u \in W^{2,p}_{\mathrm{loc}}$ with $p > n$, one has $\mathcal{H}^{n-1}(\{u = t\} \cap \{|\nabla u| = 0\}) = 0$ for a.e. $t \in (0, \|u\|_\infty)$. By the De Giorgi structure theorem, $\partial^*\{u > t\} \subset \{u = t\}$ up to an $\mathcal{H}^{n-1}$-null set for a.e. t (Maggi [3, §15]; [2] Thm. 3.59). Reparametrizing by ρ via (1) yields (4). □

Remark 3.2 (Backup citations)**.** The same statement is recorded in De Philippis–Marini–Mukoseeva 2021, Lemma 2.4, with a self-contained proof. The classical Brothers–Ziemer 1988, Lemma 2.4, gives a $W^{2,p}$ Sard statement modulo a capacity-to-$\mathcal{H}^{n-1}$ conversion.

Corollary 3.3 (V_ρ is pointwise defined and L^1)**.** *For a.e. $\rho \in [\rho_*, 1]$, V_ρ is an $\mathcal{H}^{n-1}$-a.e. finite, strictly positive Borel function on $\partial^* E_\rho$, and*

$$\int_{\partial^* E_\rho} V_\rho \, d\mathcal{H}^{n-1} = \frac{d}{d\rho}|E_\rho| = n\omega_n \rho^{n-1} =: A_\rho. \tag{5}$$

In particular $V_\rho \in L^1(\partial^* E_\rho, \mathcal{H}^{n-1})$.

4 Centre fields

Definition 4.1 (Borel centre field)**.** A *centre field* for $\{E_\rho\}$ is a Borel-measurable map $z : [\rho_*, 1] \to \mathbb{R}^n$ with $\sup_\rho |z_\rho| \le R'$ for some $R' < \infty$ that is any radius containing all translates considered. For each $\rho \in [\rho_*, 1]$ and $x \in \partial^* E_\rho$, the associated *homothetic velocity* is

$$H_{z_\rho,\rho}(x) := \frac{(x - z_\rho) \cdot \nu_\rho(x)}{\rho}. \tag{6}$$

Remark 4.2 (Downstream centre choices)**.** This building block treats z_ρ as an abstract Borel centre field. The repaired metric trace block instantiates z_ρ as a Fusco–Julin oscillation centre on good levels and sets it to a fixed bounded centre on bad levels. The centroid field of [6] remains useful for the auxiliary space–time Π identity, but it is not needed for the metric closure.

5 The metric first-variation theorem (T)

Theorem 5.1 (Metric first variation modulo translations)**.** *Under the standing hypotheses of Section 1 and for any Borel centre field $z = \{z_\rho\}$ with $\sup_\rho |z_\rho| \le R' < \infty$, the curve $\rho \mapsto [F_\rho]$ is absolutely continuous from $[\rho_*, 1]$ into $(\mathcal{X}, d_\mathcal{X})$, and there is a constant $C_{n,\rho_*} > 0$ such that*

$$|\dot{F}_\rho|_\mathcal{X} \le C_{n,\rho_*} \inf_{a\in\mathbb{R}^n} \int_{\partial^* E_\rho} \big|V_\rho - H_{z_\rho,\rho} - a \cdot \nu_\rho\big|\, d\mathcal{H}^{n-1} \tag{7}$$

for a.e. $\rho \in [\rho_*, 1]$.

5.1 Smooth flows: the moving-region identity

Suppose temporarily that Ω is smooth, $u \in C^\infty(\overline{\Omega})$, and $|\nabla u| > 0$ on the strip $\bigcup_{\rho\in[\rho_*,1-\eta]} \partial E_\rho$ for some $\eta > 0$. The homothety $\Psi_\rho(y) := z_\rho + \rho^{-1}(y - z_\rho)$ maps E_ρ onto $z_\rho + \rho^{-1}(E_\rho - z_\rho)$.

Lemma 5.2 (Pulled-back normal velocity)**.** *In the smooth case, the normal velocity of the rescaled-and-translated flow $\rho \mapsto F_\rho - \rho a$, pulled back to ∂E_ρ via Ψ_ρ, equals*

$$W_\rho^{z_\rho,a} := \rho^{-1}\big(V_\rho - H_{z_\rho,\rho} - a \cdot \nu_\rho\big) \qquad \text{on } \partial E_\rho. \tag{8}$$

Proposition 5.3 (Smooth case of (T))**.** *In the smooth setting above,*

$$\limsup_{h\to 0} \frac{d_\mathcal{X}([F_{\rho+h}], [F_\rho])}{|h|} \le \rho_*^{-n} \inf_{a\in\mathbb{R}^n} \int_{\partial^* E_\rho} \big|V_\rho - H_{z_\rho,\rho} - a \cdot \nu_\rho\big|\, d\mathcal{H}^{n-1}. \tag{9}$$

The proof is the first-variation identity [3, §17.3] applied to $\widetilde{F}_\rho := F_\rho - \rho a$, followed by the homothety Jacobian (factor ρ^{1-n}) and the translation infimum.

5.2 BV/coarea approximation and finite-perimeter passage

Lemma 5.4 (Coarea-good smooth recovery)**.** *There is a smooth nested recovery family $\{E_\rho^k\}_{\rho\in[\rho_*,1]}$ obtained by local regularisation of the torsion subgraph and monotone volume reparametrisation such that, after passing to a subsequence, for a.e. $\rho \in [\rho_*, 1]$,*

$$E_\rho^k \to E_\rho \text{ in } L^1, \qquad t_k(\rho) \to t(\rho), \qquad P(E_\rho) \le \liminf_{k\to\infty} P(E_\rho^k).$$

Moreover the corresponding normal-velocity measures converge in the weak strict BV sense needed for Reshetnyak lower semicontinuity in Lemma 5.6.

Proof (sketch). Mollification of u followed by coarea-good level selection produces the nested smooth family; the BV/coarea machinery ([3, §13]) gives L^1 convergence of a.e. regular level together with perimeter lower semicontinuity, and the Reshetnyak setup of [3, §17] (see also [2, Thm. 2.39]) supplies the weak strict BV convergence of the normal-velocity measures required in Lemma 5.6. $\square$

Remark 5.5 (Why this formulation is weaker and correct)**.** The recovery statement is not an arbitrary mollification claim, and it does not assert pointwise perimeter convergence for every level. It is the standard coarea-good relaxation used for finite-perimeter first variations: choose regular levels outside a null set, regularise locally away from the critical set supplied by Figalli–Maggi, and keep only the lower semicontinuity needed to pass the smooth estimate to the reduced boundary.

Lemma 5.6 (Reshetnyak lower semicontinuity)**.** *For every $z \in \mathbb{R}^n$ and $a \in \mathbb{R}^n$, the convex linear-growth integrand $\Phi(x, s, \nu) := |s - (x - z) \cdot \nu/\rho - a \cdot \nu|$ satisfies, for a.e. $\rho \in [\rho_*, 1]$,*

$$\liminf_{k\to\infty} \int_{\partial^* E^k_\rho} \big|V^k_\rho - H_{z,\rho} - a \cdot \nu^k_\rho\big|\, d\mathcal{H}^{n-1} \geq \int_{\partial^* E_\rho} \big|V_\rho - H_{z,\rho} - a \cdot \nu_\rho\big|\, d\mathcal{H}^{n-1}. \tag{10}$$

Since Φ is continuous in x on the support, the x-dependent Reshetnyak lower semicontinuity of [2, Thm. 2.39] applies.

Lemma 5.7 (AGS lower semicontinuity of the metric derivative)**.** *Let $\gamma_k : [\rho_*, 1] \to \mathcal{X}$ be AC curves with $\gamma_k(\rho) \to \gamma(\rho)$ in $\mathcal{X}$ for every ρ, and $|\dot\gamma_k|(\rho) \leq m_k(\rho)$ a.e. with $m_k \rightharpoonup m$ weakly in $L^1([\rho_*, 1])$. Then γ is AC and $|\dot\gamma|(\rho) \leq m(\rho)$ for a.e. $\rho \in [\rho_*, 1]$. ([1] Thm. 1.1.2)*

Proof of Theorem 5.1. By Lemma 5.4, after extracting a subsequence we may assume $E^k_\rho \to E_\rho$ in L^1 for a.e. $\rho \in [\rho_*, 1]$. By Proposition 5.3 applied to the smooth approximating flow $\{E^k_\rho\}$,

$$|\dot F^k_\rho|_{\mathcal{X}} \leq m_k(\rho) := \rho_*^{-n} \inf_{a\in\mathbb{R}^n} \int_{\partial^* E^k_\rho} \big|V^k_\rho - H_{z_\rho,\rho} - a \cdot \nu^k_\rho\big|\, d\mathcal{H}^{n-1}. \tag{11}$$

The coarea-good recovery gives uniform integrability of $\{m_k\}$ on $[\rho_*, 1]$, so by Dunford–Pettis it admits a weakly L^1-convergent subsequence $m_k \rightharpoonup m$. By Lemma 5.6, $m(\rho) \geq \rho_*^{-n} \inf_a \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho,\rho} - a \cdot \nu_\rho|$ for a.e. ρ. By Lemma 5.7, $\rho \mapsto [F_\rho]$ is AC and $|\dot F_\rho|_{\mathcal{X}} \leq m(\rho)$ a.e. This is (7) with $C_{n,\rho_*} = \rho_*^{-n}$. $\square$

6 Translations quotient kills the $a \cdot \nu_\rho$ mode

Lemma 6.1 (Translation derivative formula)**.** *For every bounded set $E \subset \mathbb{R}^n$ of finite perimeter and every $a \in \mathbb{R}^n$,*

$$\lim_{\varepsilon\to 0} \frac{|E\Delta(E + \varepsilon a)|}{\varepsilon} = \int_{\partial^* E} |a \cdot \nu_E|\, d\mathcal{H}^{n-1}. \tag{12}$$

([3] Thm. 13.8; [2] Thm. 3.84.)

7 Per-ρ squared form: status and open issues

7.1 Pointwise (per-ρ) version: conditional

If one assumes the per-ρ radial trace bound (statement (R), to be supplied by Building Block [8]; audit-trail markdown note) and the homothetic velocity defect bound,

$$\Big(\int_{\partial^* E_\rho} |H_{z_\rho,\rho} - \bar V_\rho|\, d\mathcal{H}^{n-1}\Big)^2 \leq C_{n,\rho_*,R} D_I(t(\rho)), \tag{13}$$

then Theorem 5.1, the triangle inequality, and Cauchy–Schwarz combine to give the pointwise per-ρ squared bound

$$|\dot{F}_\rho|^2_{\mathcal{X}} \le C_{n,\rho_*,R}\big(D_H(t(\rho)) + D_I(t(\rho))\big). \tag{14}$$

Remark 7.1 (Honest status)**.** The estimate (13) is conditional on the pointwise (R), which is open in the no-graph setting. Consequently (14) is *conditional* and not used downstream.

7.2 Integrated version: repaired route

The Plan 2 paper closes the bound only in its μ-integrated good/bad form. This integrated form is established downstream using Theorem 5.1 as input; see [8], where Theorem 5.1 is combined with the levelwise Fusco–Julin centre on small-isoperimetric-defect levels, the oriented radial trace estimate, and the uniform Lipschitz bound on the large-defect levels, yielding the μ-integrated metric-energy bound without per-ρ (R) and without the centroid integrated Π input.

8 Summary and use downstream

Theorem 8.1 (Statement of MetricFramework)**.** *Let $n \ge 2$, $0 < \rho_* < 1$, and $R \ge 2$. Let $\Omega \subset B_R$ be bounded open with $|\Omega| = \omega_n$, let u be its torsion potential, let $\{E_\rho\}$ be the volume-radius foliation* (1)*, and let $F_\rho := \rho^{-1}E_\rho$ in $\mathcal{X}$. For every Borel centre field $z = \{z_\rho\}$ with $\sup_\rho |z_\rho| \le R'$:*

- *the curve $\rho \mapsto [F_\rho] \in \mathcal{X}$ is absolutely continuous on $[\rho_*, 1]$;*
- *the normal velocity $V_\rho = -t'(\rho)/|\nabla u|$ is $\mathcal{H}^{n-1}$-a.e. finite and positive on $\partial^* E_\rho$ for a.e. ρ;*
- *there is a constant $C_{n,\rho_*} > 0$ such that*

$$|\dot{F}_\rho|_{\mathcal{X}} \le C_{n,\rho_*} \inf_{a\in\mathbb{R}^n} \int_{\partial^* E_\rho} \big|V_\rho - H_{z_\rho,\rho} - a\cdot\nu_\rho\big|\, d\mathcal{H}^{n-1} \qquad \text{for a.e. } \rho \in [\rho_*, 1].$$

References

[1] L. Ambrosio, N. Gigli, G. Savaré, *Gradient Flows in Metric Spaces and in the Space of Probability Measures*, 2nd ed., Birkhäuser, 2008.

[2] L. Ambrosio, N. Fusco, D. Pallara, *Functions of Bounded Variation and Free Discontinuity Problems*, Oxford, 2000.

[3] F. Maggi, *Sets of Finite Perimeter and Geometric Variational Problems*, Cambridge, 2012.

[4] A. Figalli, F. Maggi, ARMA **201** (2011), 143–207. App. A, Thm. A.1.

[5] *Building Block* LevelSetIdentity, Plan 2.

[6] *Building Block* CentroidBound, Plan 2.

[7] *Building Block* SpaceTimeIdentity, Plan 2.

[8] *Building Block* WeightedMetricTrace, Plan 2.